Ritwick Verma

Backend Engineer · C++ · Search Systems
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SUMMARY

Backend engineer with 5.5 years on Cassini, the distributed C++ search engine behind every ebay.com search (1B+ listings). Owned eBay Motors fitment; built caching and query transforms on the serving path. 215 merged PRs.

EXPERIENCE

Software Engineer 3, Search Backend, eBay, San Francisco, CA — Jun 2023 - Present

- **Owned eBay Motors fitment**, the C++ subsystem that decides which parts fit a buyer's vehicle on 16 sites and 19 catalogs. Top author of its current code (31%) and point of contact for the partner teams that rely on it.
- **Led fitment's move to the engine's new transform architecture**, then moved its production index and test fixtures off a legacy query language with no cutover outage, keeping all 2,549 unit and 511 regression tests green.
- **Replayed 50,000 production-sampled queries before merging my refactor** of the client fitment operator, traced the new failures to init-order-dependent shared config and fixed the root cause: replay errors 167 to 6 (fix in review).
- **Built a request-scoped cache** for recall recovery that needs no invalidation and never caches transient failures. Also in recall: client-supplied vectors skip the embedding service, and big requests run as concurrent batches.
- **Built three C++ query transforms** for results-page product carousels (18 PRs): one batched round trip gets each product's listing count and lowest price under the user's filters; a failed table is dropped, not shown half-built.
- **Designed the engine side of a new fitment category** (pre-launch): showed a proposed bitmask match accepted packs with the wrong piece counts, kept vehicle matching on the proven pipeline, raised a requirements conflict early.
- **Wrote the Markdown-driven test frameworks** (tests as readable tables) the team made its standard. Made the tokenizer stack allocator-aware (C++ PMR) and served it as an internal HTTP API that science and QA use.

Software Engineer 2, Search Backend, eBay, San Jose, CA — Mar 2021 - Jun 2023

- **Took over fitment in year two** (code 6 to 13 years old, no unit tests): modernized its memory handling, refactored its worst parts, wrote its first unit tests, and rebuilt the config generators so config stopped waiting on engine releases.
- **Built garage-vehicle fitment** in a cross-team Motors effort: constraints come from the buyer's vehicle, cutting two service calls from the latency path. Made tire fitment-by-spec config-driven, so new countries need no engine change.
- **Co-built fitment's A/B testing** (control and treatment on one request, only affected queries tagged). Fixed a production memory spike by capping a combinatorial expansion. Moved core request containers to PMR allocators.

Earlier: software intern at Shoelace Wireless (Android, 2020) and BHEL (Java EE, 2017).

PROJECTS

Built with AI coding agents (Hermes, self-hosted) under my direction. Also ShutUp, an Android app with 26,000+ installs.

- **TizenAmbilight:** Ambilight from inside a Samsung TV. The public capture API only returns the UI layer, so it reads the undocumented video plane instead, tone-maps HDR and drives an LED strip at 20 to 25 fps.
- **Plutus:** Ran 30+ trading-strategy experiments on point-in-time data with bootstrap confidence intervals; none held up.
- **Homelab as code:** 16 Proxmox guests, 14 declared in OpenTofu, with a nightly CI job that flags config drift.

SKILLS

- **Languages:** C++ (17/20/23), Python, Java
- **Search:** Query understanding (rewriting, tokenization, spell correction), ranking factors, embedding retrieval, caching
- **Tooling:** C++ PMR, ASan, clang-tidy, CMake, GoogleTest, Git, Jenkins, Linux, A/B tests, traffic replay, Claude Code

EDUCATION

Master of Computer Science, University of California, Irvine — 2019 - 2020
B.Tech, Computer Science and Engineering, SRM Institute of Science and Technology — 2015 - 2019